A Multi-Agent Academic Helpdesk for IIT Jodhpur

Assignment 2: Code, Design, Evaluation and Findings

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Abstract

This project implements a local academic helpdesk that retrieves institutional documents, answers with citations, looks up course information, drafts emails for review and records unanswered requests. An orchestrator delegates to policy, schedule and email roles using Ollama and Chroma. Evaluation uses 25 questions, repeated runs and a separate comparison of Qwen3 1.7B and Llama 3.2 3B. Findings show useful source-backed answering and consistent refusal on the tested out-of-scope requests, alongside incomplete schedule integration, unsupported counting and citation-applicability concerns.

1 Problem statement

The helpdesk must answer academic programme, curriculum, attendance, registration and examination questions. It should look up slots and instructor contacts, draft sick-leave or makeup-exam emails without sending them, cite evidence and log questions requiring human help. Unrelated requests should receive an unknown response rather than a guess. The assignment also requires a document corpus, single-agent and multi-agent implementations, and evaluation on 20–30 prompts.

2 System design

Question → clarification checks → orchestrator

Policy role → document retrieval → cited answer
Schedule role → CSV lookup or guarded offerings count
Email role → instructor lookup → unsent draft

Unsupported request → unanswered-question log

Policy and email roles use different prompts with the same local chat model. Schedule lookup uses deterministic tools. All roles run in one Python process; Google ADK is not used. Multi-agent mode is the default. Single-agent mode uses shared routing, retrieval and tools with a common role prompt, and is therefore a limited architectural baseline.

The default chat model is `qwen3:1.7b`, with temperature zero and a 2,048-token context. `nomic-embed-text` generates embeddings. The smaller chat model accommodates the available 8 GB machine, but peak total memory has not been measured and the assignment's 16 GB target is not formally profiled.

2.1 Retrieval and document handling

Text is normalized and split into 1,100-character chunks with 200-character overlap. Chroma stores vectors and performs cosine-based retrieval. Retrieved evidence supports answers with

source titles, URLs and PDF pages or Office document locations. PDF, HTML, DOCX and XLSX are supported. Excel extraction uses saved cell values; formula-based workbooks require saved calculated results.

Scanned PDF pages can use OCR sidecars matched to the original PDF by SHA-256. OCR-derived citations are marked for checking. Full ingestion builds a replacement collection before switching it live, preserving the previous collection if an embedding build fails. Selective updates remove that source's old chunks first.

2.2 Tools and controls

Timetable and instructor tools read CSV files with semester context. Counting requires a complete verified offerings list. Email drafting uses placeholders for missing details and never sends messages. Unknown requests are recorded in `logs/unanswered.jsonl`. Citation validation checks referenced evidence numbers, but does not prove semantic support or programme applicability.

3 Corpus and structured data

The main corpus has 54 registered records: 22 HTML documents, 24 PDFs, one Word document and seven Excel workbooks, producing 3,922 indexed chunks. It covers regulations, amendments, curricula, calendars, registration, faculty information and schedules. The manifest records provenance; local attachments retain local file citations. The main evaluation uses all 54 sources.

Structured coverage is narrower. The timetable CSV contains three first-year entries for 2023–24 Semester-I, and the instructor CSV contains one CSL7570 assignment for May 2022. The offerings completeness declaration is empty. The Semester I AY 2026–27 workbook lists CSL7650 Autonomous Systems, slot E, Gaurav Harit, room LHC 106, at Classroom Allocation row 150. The finalized Semester I list agrees at CSE row 26. These spreadsheet rows are not imported automatically into the CSV tools. CSL7650 and CSL7570 are different courses. Humanities counts require reconciliation of department names, levels, elective categories and semester coverage before an exhaustive count is justified.

4 Evaluation methodology

The 25 questions comprise 12 policy prompts, two clarification cases, four missing-data cases, two email drafts and five out-of-scope requests. They are provided in `TEST_QUESTIONS.md` and `evaluation/questions.json`. Assessment considers reference-answer agreement, support for cited claims, correct out-of-scope refusals and substantive changes between repeated runs. Drafts are checked for required placeholders and invented details.

Reference-answer agreement includes clarification, refusal and drafting behavior, rather than factual answering alone. Groundedness considers cited policy answers, excluding uncited refusals and drafts; high grounding can coexist with low coverage. The questions were also used during development, so results do not establish held-out general accuracy.

The main evaluation consists of two Qwen multi-agent runs over the 54-source corpus. A separate comparison uses two runs per model over a 30-source, 860-chunk configuration. Its model scores do not measure the main configuration.

5 Results and findings

5.1 Main evaluation

Measure	Run 1	Run 2
Matches reference expectations	23/25 (92%)	22/25 (88%)
Correct out-of-scope refusals	5/5	5/5
Runtime errors	0	0

The combined reference agreement is 45/50, with important interpretation limits. Question 10 expects an unknown response for Autonomous Systems despite its assignment existing in the spreadsheets; the CSV tool still cannot return it. Question 1 gives the expected numeric credit requirement but cites a different curriculum revision. Agreement with the reference therefore overstates current coverage and does not establish citation applicability.

All ten out-of-scope requests are refused. Question 3 correctly gives 8 May as the summer class start. Questions 22 and 25 remain unanswered despite calendar evidence for the second- and third-fractal start dates, 19 May and 30 May.

Question 11 changes from refusal to an unsupported count of 12 Humanities electives. Policy routing bypasses the schedule completeness guard and confuses curriculum credits with offered courses. A valid citation identifier does not support the claim. Because of this failure and the curriculum-applicability concern, no aggregate grounding percentage is assigned to the main evaluation.

5.2 Model comparison

Measure	Qwen3 1.7B	Llama 3.2 3B
Appropriate behavior, run 1	21/25	13/25
Appropriate behavior, run 2	22/25	13/25
Appropriate behavior, combined	43/50 (86%)	26/50 (52%)
Supported cited policy answers	17/18	2/2
Correct out-of-scope refusals	10/10	10/10
Median end-to-end time	2.95 s	0.63 s
Observed loaded chat-model size	1.6 GB	2.3 GB

Qwen answers more policy questions in this configuration. Llama frequently refuses answerable questions and misroutes a curriculum question to timetable handling. Its makeup-exam draft invents treatment and recovery details. Llama's 2/2 grounding result covers only two cited answers and does not imply good coverage. Its lower median includes early refusals and is not a fair generation-speed comparison. Timings include routing, retrieval and generation; loaded-model size is not peak system RAM. Unequal model sizes and Qwen-oriented application tuning limit broader conclusions.

5.3 Repeated-run variation

Main-evaluation questions 2, 3, 6, 11, 13, 23 and 24 change text. Question 11 changes substantively from refusal to an unsupported count; other differences concern wording, formatting or source choices. In the model comparison, Qwen's question 24 changes from the wrong fee

deadline of 13 June to the correct 13 May. Llama's question 13 changes draft wording, while invented details in question 14 persist. Temperature zero does not guarantee identical responses.

6 Conclusion

The prototype demonstrates local retrieval, delegation, cited answers, dated lookup tools, unsent drafts and unanswered logging. Qwen provides better answer coverage than Llama on the recorded comparison. Both refuse all tested out-of-scope requests.

Remaining limitations include spreadsheet-to-CSV integration, a count guard that routing can bypass, exact semester-label matching, missing current contacts, OCR errors and insufficient citation-support and cohort-applicability checks. The small development set and unmeasured peak memory limit general claims. A matched, repeated and scored single-versus-multi experiment is needed before claiming an accuracy gain from delegation. Important academic answers and email drafts require human review.